Analysis of Algorithms
Lab 07

1. Implement the forward elimination algorithm for the Gaussian elimination.
2. Design the backwards substitution algorithm and show that its running time is in $\Theta(n^2)$.
3. Implement an algorithm to transform a random array into a heap.
4. Implement an algorithm that can solve a system of equations using LU decomposition. LU decomposition has to be based on Gaussian Elimination.

Submission

Upload your solutions as a PDF file. In case of coding assignments, upload to GitHub and include the link in the PDF file. Your submission should include screenshots of the execution of your code when applicable.